Lesson 03 Demo 02

Creating a Form with Three Input Fields in React

Objective: To develop a form with three input fields: Name, Email, and Message using React with Vite for a faster and optimized development experience

Tools required: Visual Studio Code and Node.js

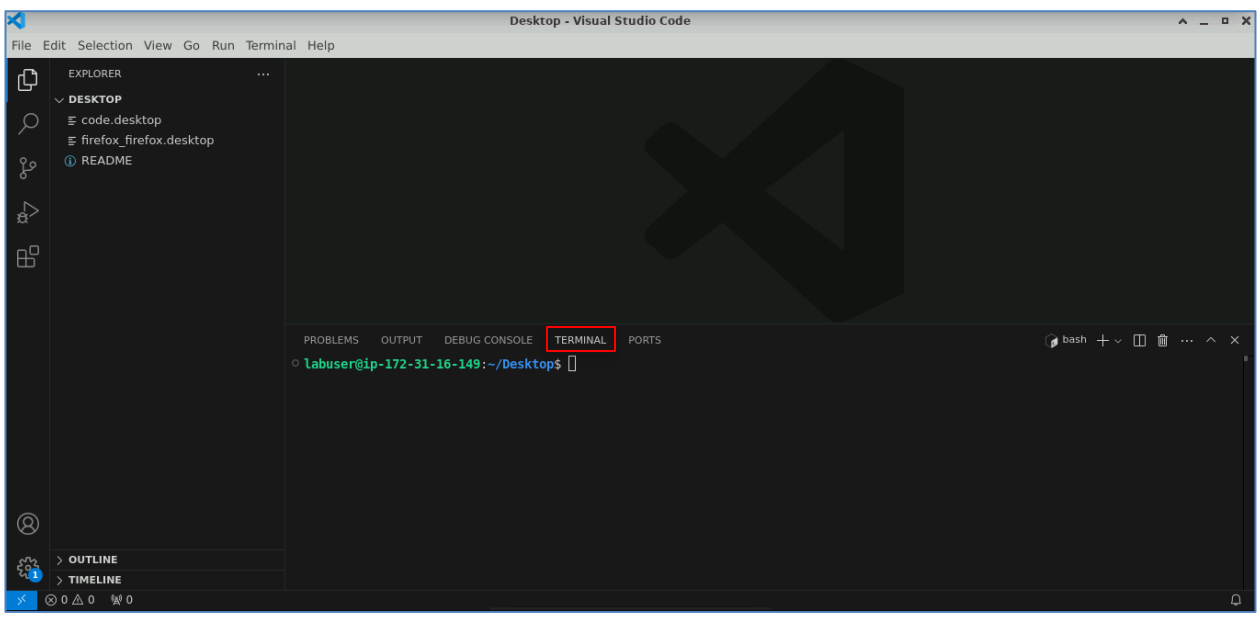
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create a React component
3. Import and use the component
4. Run and verify the application

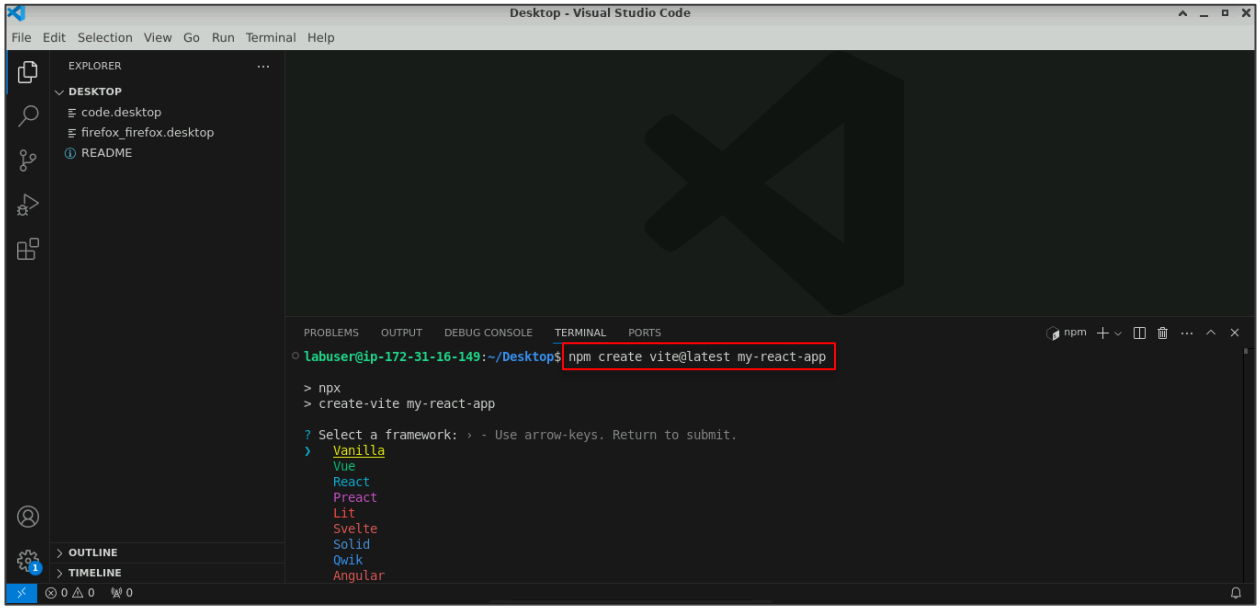
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

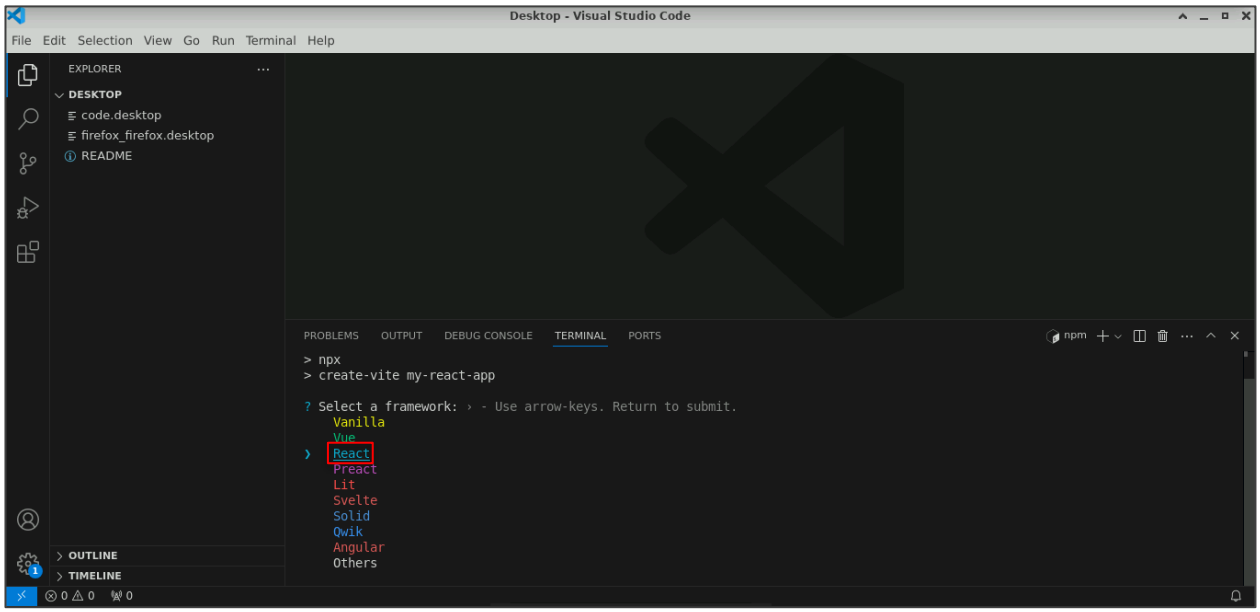


1.2 Run the following command to create a new React project using Vite:

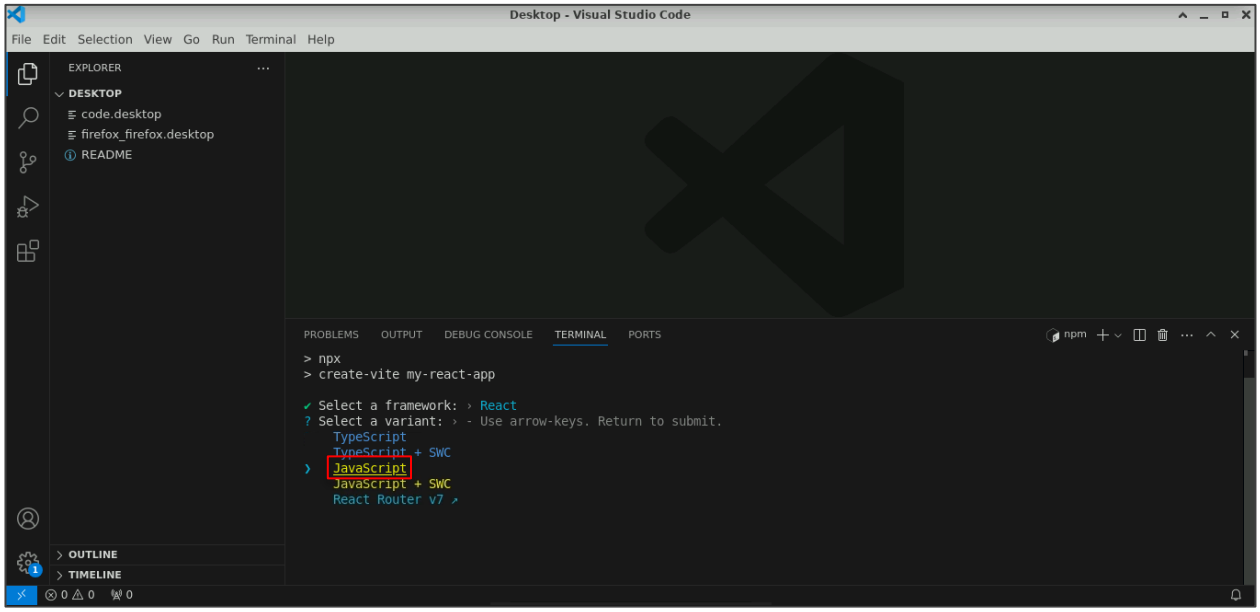
npm create vite@latest my-react-app



1.3 Select **React** and press enter

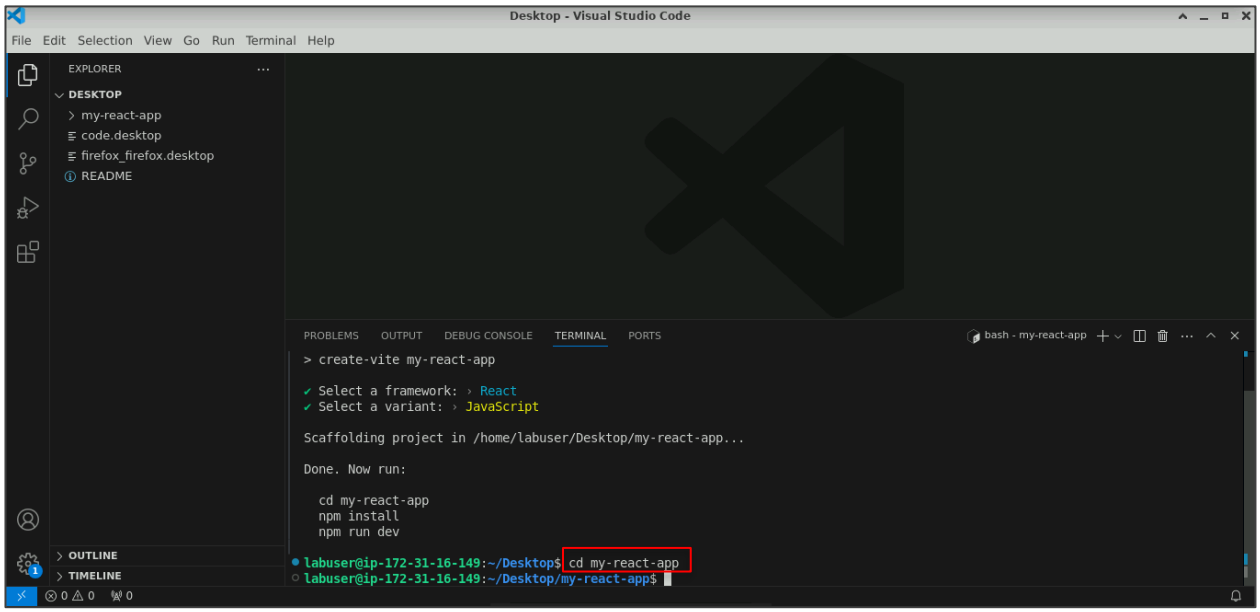


1.4 Select **JavaScript** and press enter



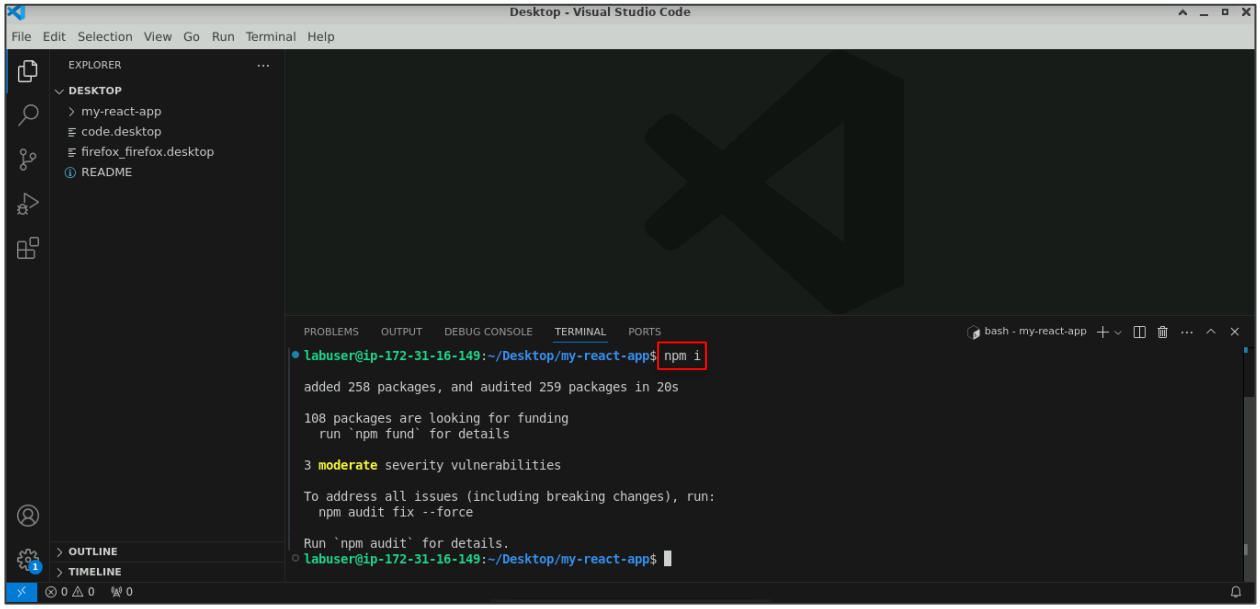
1.5 Run the following command to navigate to the project folder:

cd my-react-app



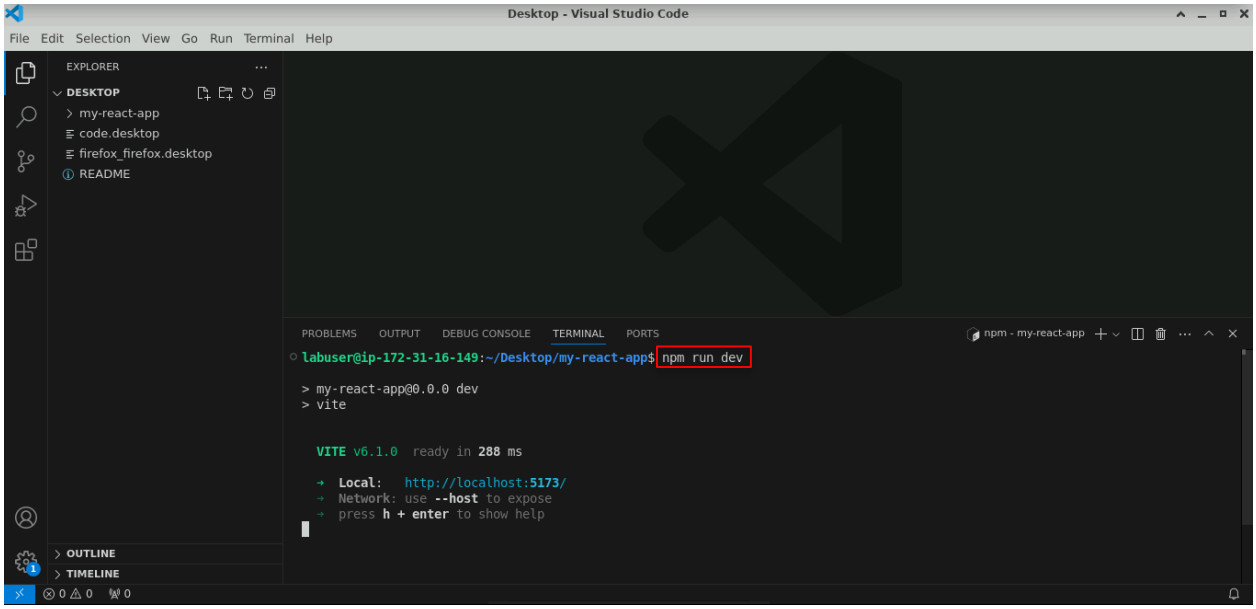
1.6 Run the following command to install dependencies:

npm i



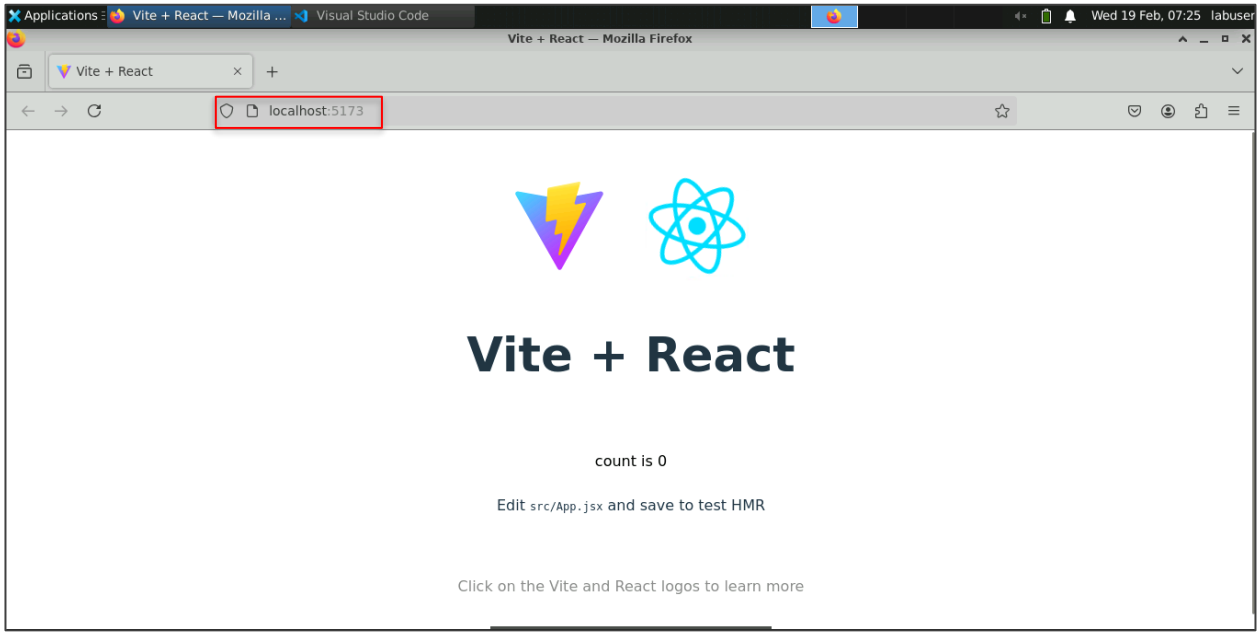
1.7 Run the following command to start the development server:

npm run dev



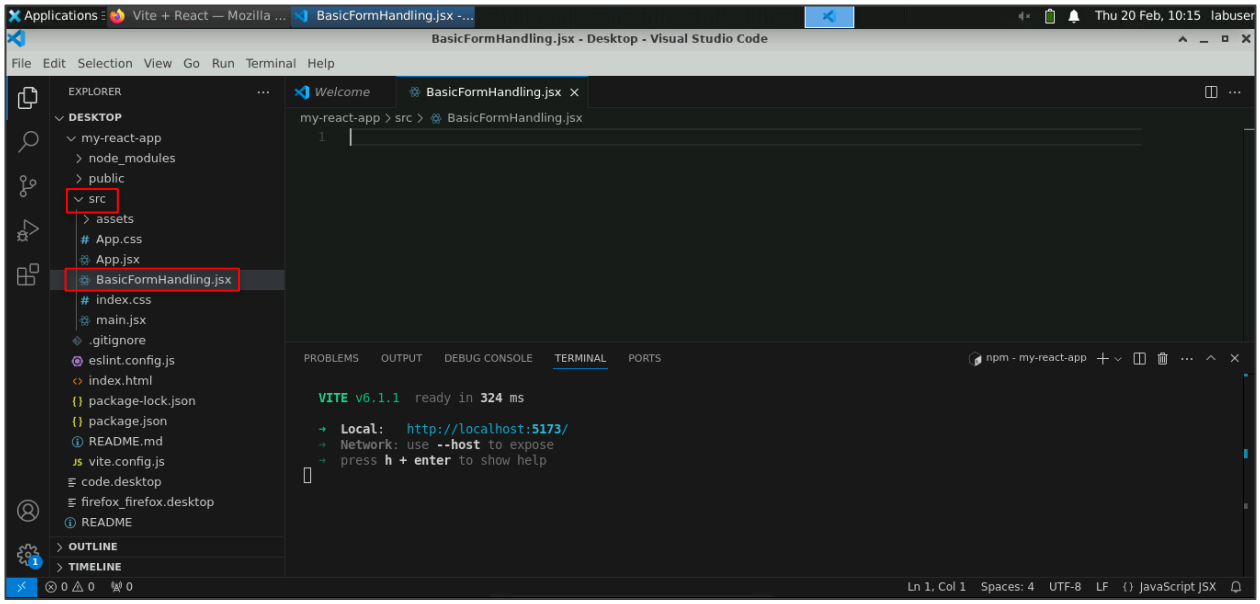
1.8 Navigate to the web browser and paste the link below:

http://localhost:5173



Step 2: Create a React component

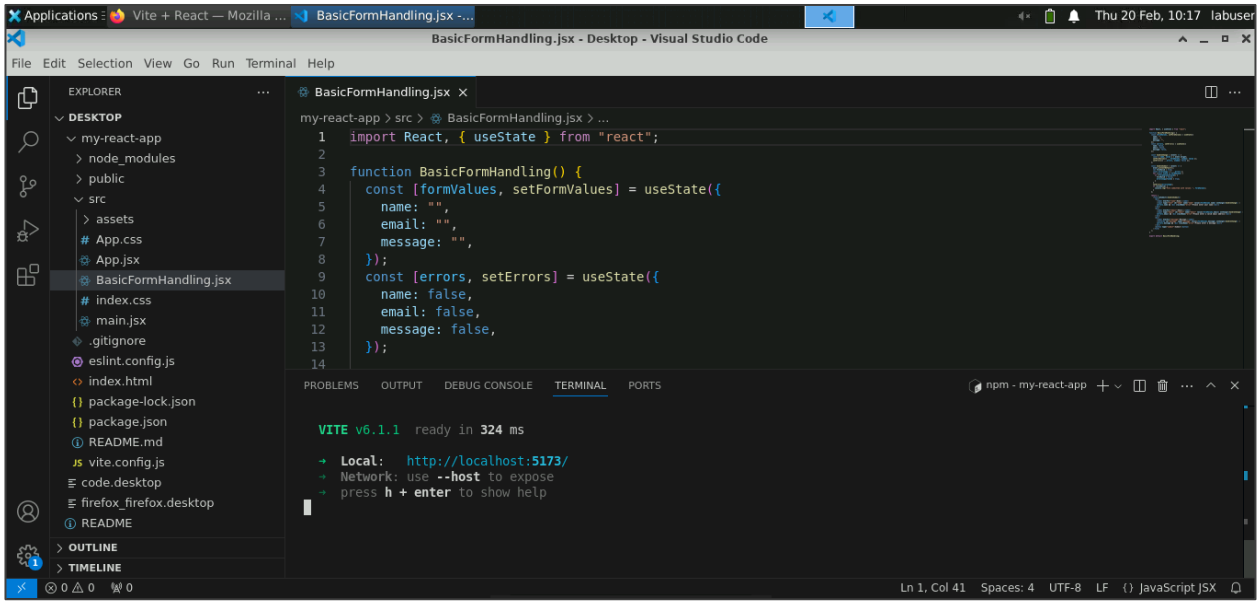
2.1 Inside the **src** folder, create a new file named **BasicFormHandling.jsx**



2.2 Add the following code to **BasicFormHandling.jsx**

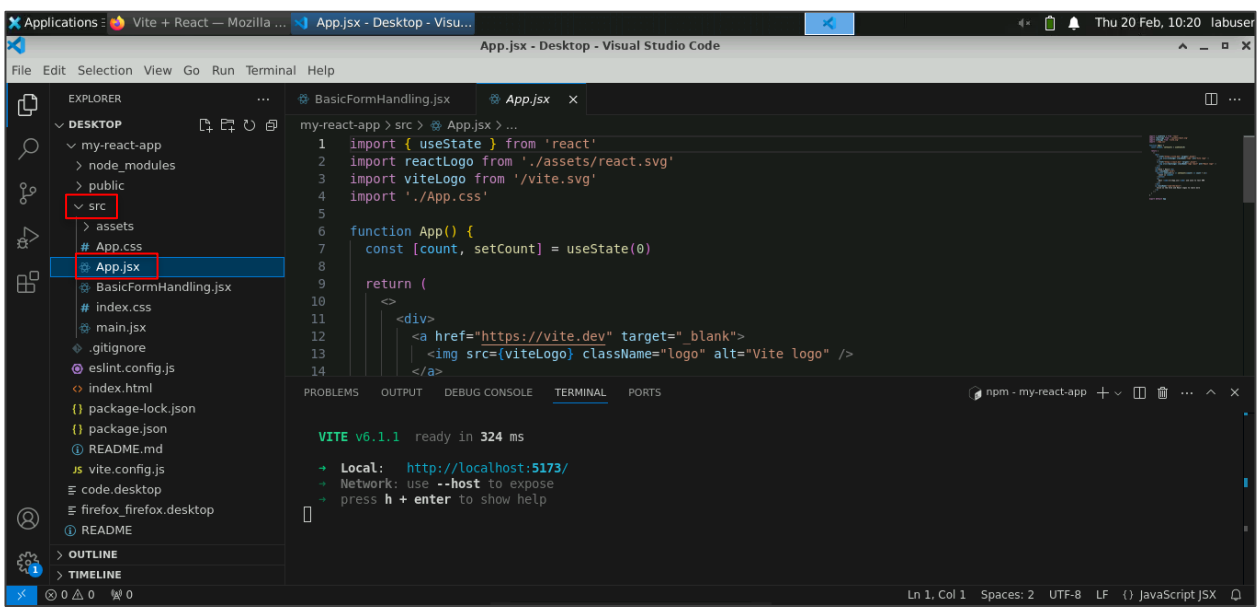
```
import React, { useState } from "react";

function BasicFormHandling() {
  const [formValues, setFormValues] = useState({
    name: "",
    email: "",
    message: "",
  });
  const [errors, setErrors] = useState({
    name: false,
    email: false,
    message: false,
  });

  const handleChange = (event) => {
    const { name, value } = event.target;
    setFormValues({ ...formValues, [name]: value });
    setErrors({ ...errors, [name]: false });
  };

  const handleSubmit = (event) => {
    event.preventDefault();
    let formValid = true;
    const errorsCopy = { ...errors };
    for (const field in formValues) {
      if (!formValues[field]) {
        formValid = false;
        errorsCopy[field] = true;
      }
    }
    setErrors(errorsCopy);
    if (formValid) {
      console.log("Form submitted with values: ", formValues);
    }
  };
```

```
  return (
    <form onSubmit={handleSubmit}>
      <div>
        <label htmlFor="name">Name:</label>
        <input type="text" id="name" name="name" value={formValues.name}
onChange={handleChange} />
        {errors.name && <div className="error">Please enter your name</div>}
      </div>
      <div>
        <label htmlFor="email">Email:</label>
        <input type="email" id="email" name="email" value={formValues.email}
onChange={handleChange} />
        {errors.email && <div className="error">Please enter a valid email
address</div>}
      </div>
      <div>
        <label htmlFor="message">Message:</label>
        <textarea id="message" name="message" value={formValues.message}
onChange={handleChange} />
        {errors.message && <div className="error">Please enter a message</div>}
      </div>
      <button type="submit">Submit</button>
    </form>
  );
}

export default BasicFormHandling;
```



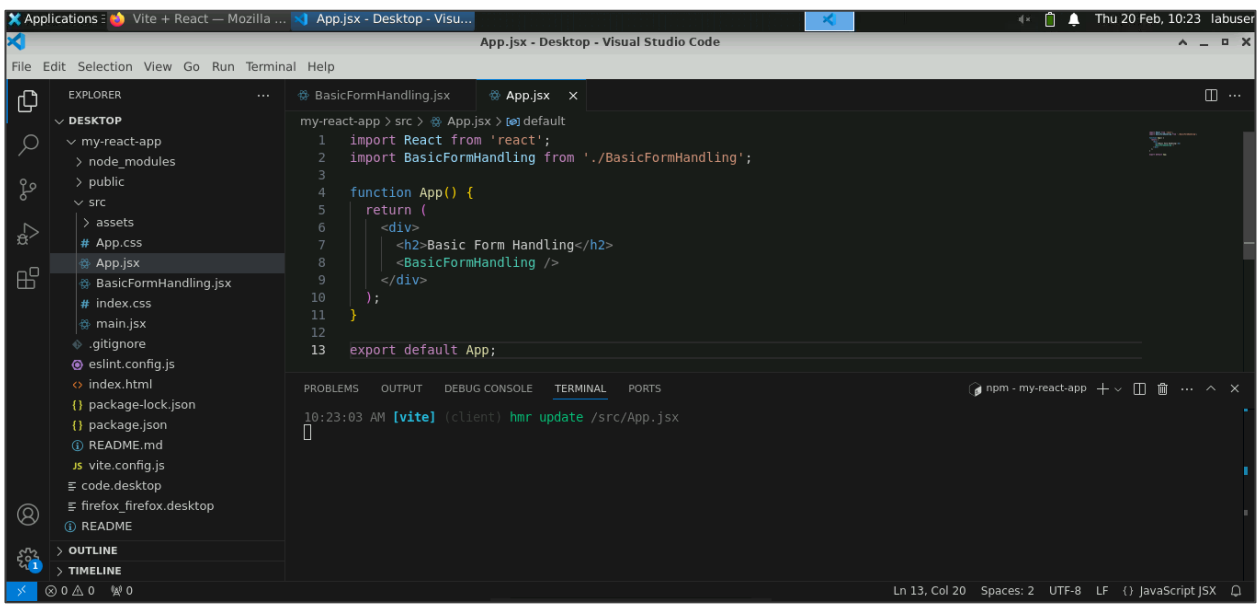
Step 3: Import and use the component

3.1 Open the **App.jsx** file inside the **src** folder



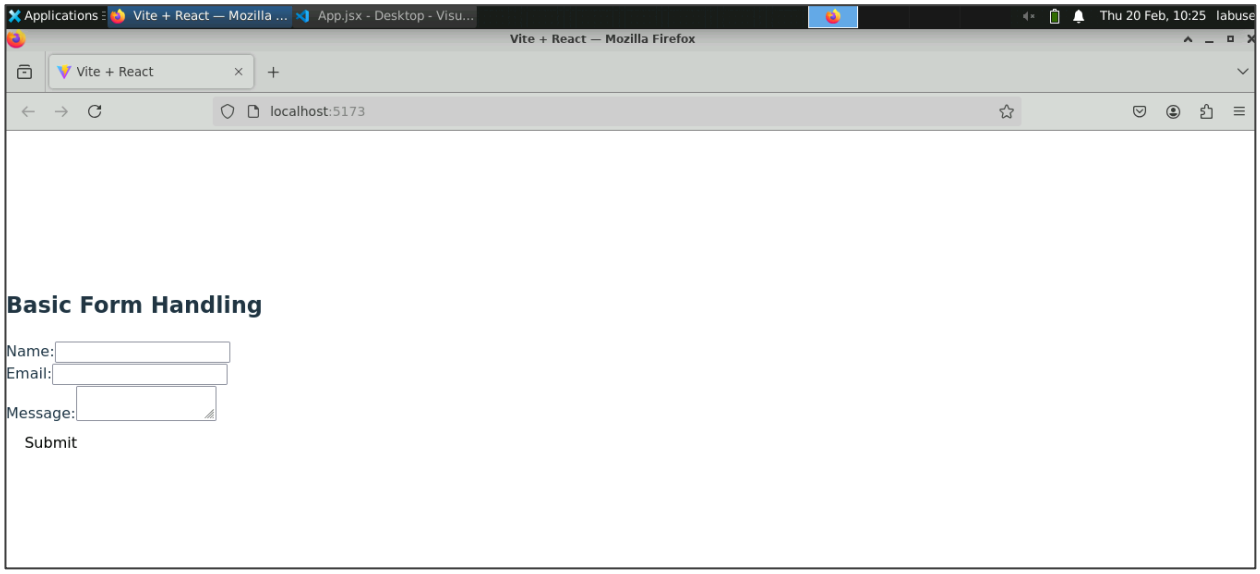
3.2 Paste the code below to include the new component in **App.jsx**:

```
import React from 'react';
import BasicFormHandling from './BasicFormHandling';

function App() {
  return (
    <div>
      <h2>Basic Form Handling</h2>
      <BasicFormHandling />
    </div>
  );
}

export default App;
```



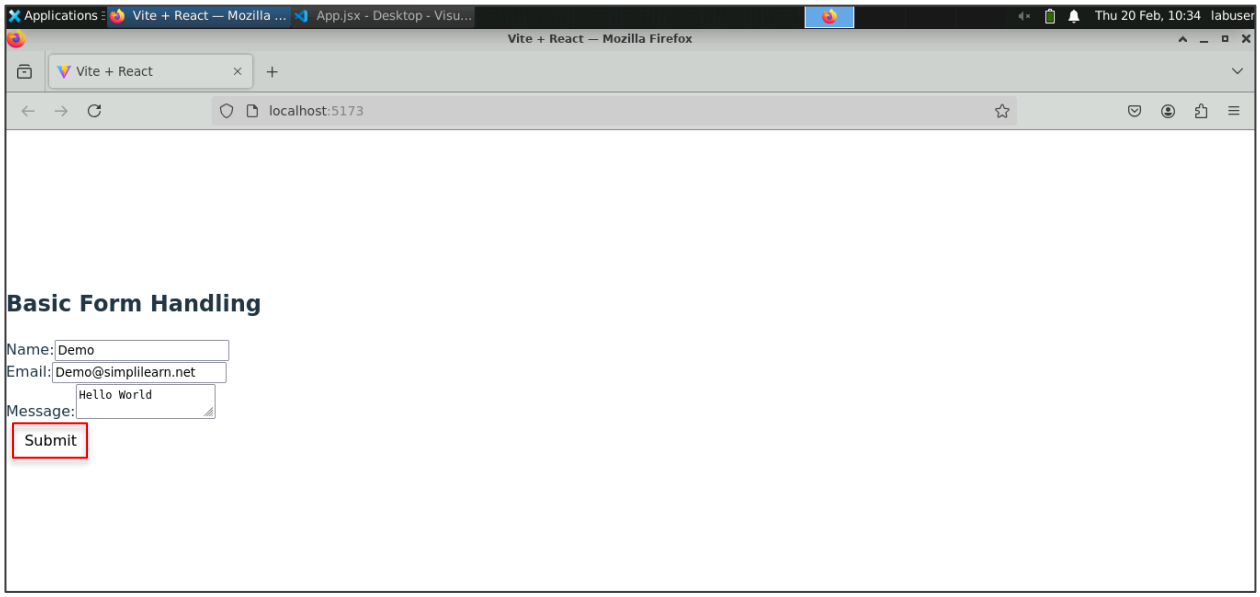
Step 4: Run and verify the application

4.1 Navigate to the browser and refresh the webpage

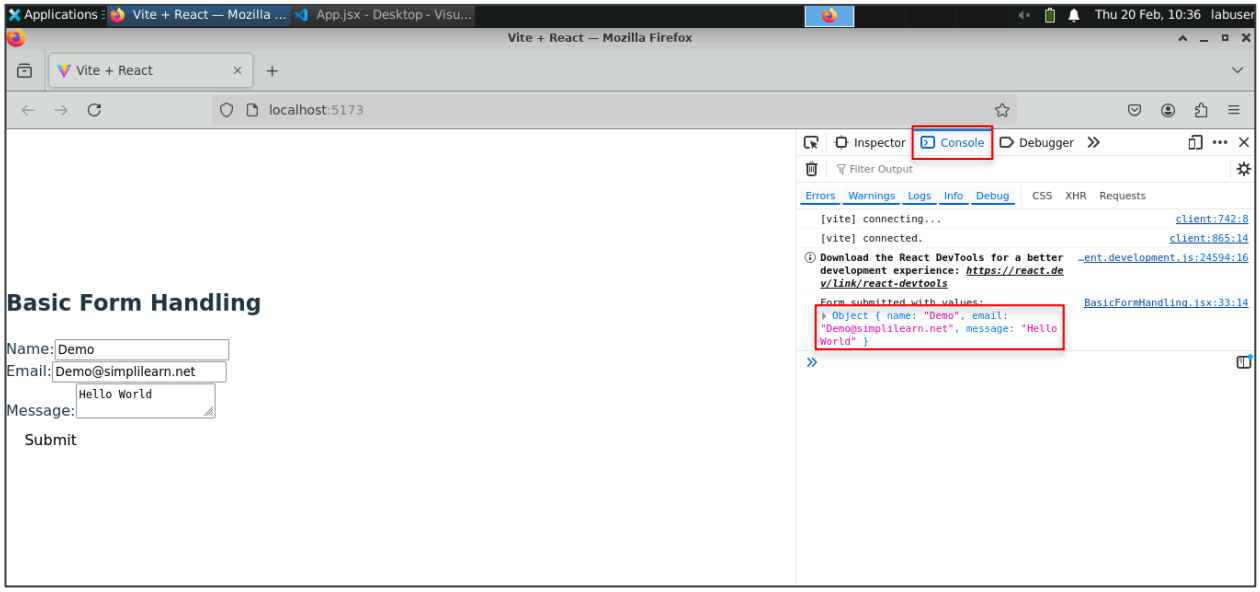


Now, you will see the basic form displayed on the screen.

4.2 Enter your credentials and click **Submit**



4.3 Press **ctrl + shift + I** to view the Console tab



You can view the details entered in the form displayed on the console.

By following the above steps, you have successfully created a React form with Name, Email, and Message fields using Vite. The form includes real-time validation and logs submitted data to the console.